

Recap:

def: - set, elements. $a \in A$

- subsets: $A \subseteq B$ all $a \in A$ and $a \in B$

- infinite sets: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$

$$\mathbb{Q} = \left\{ x : x = \frac{m}{n}, \text{ for } m \in \mathbb{Z}, n \in \mathbb{N} \right\}$$

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

- finite sets $A = \{a_1, a_2, \dots, a_n\}$

$$|A| = n$$

- when not subset: $A \not\subseteq B$ when there is $a \in A$

such that $a \notin B$.

Fact: $\emptyset = \{\} \subseteq A$ for any set A .

- operations:

$$\text{union: } A \cup B = \{x : x \in A \text{ or } x \in B\}$$

$$\text{intersection: } A \cap B = \{x : x \in A \text{ and } x \in B\}$$

$$\text{set difference: } A - B = \{x : x \in A \text{ and } x \notin B\}$$

complement: $\bar{A} = U - A = \{x : x \in U \text{ and } x \notin A\}$
 $= \{x \in U : x \notin A\}$

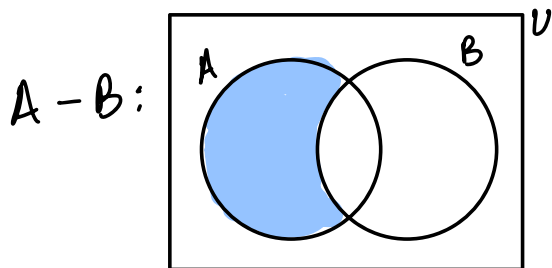
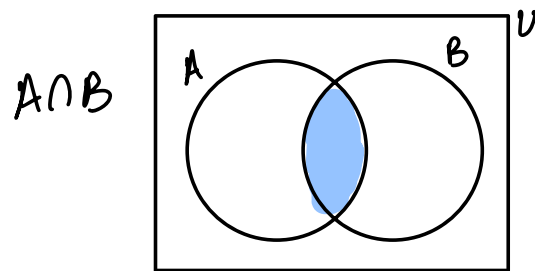
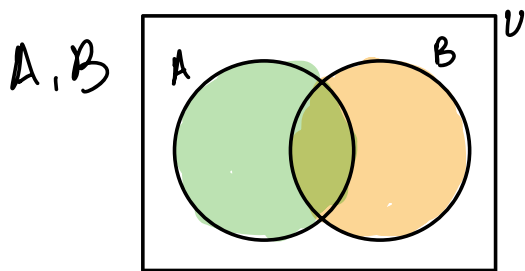
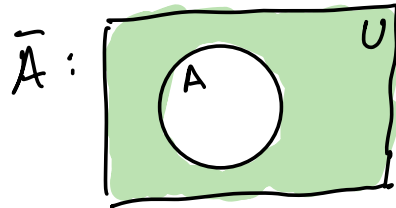
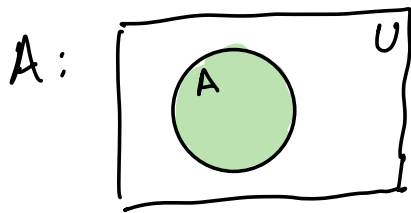
if $A = \{-2, 1, 0\} \subseteq \mathbb{Z}$

$B = \{0, \pi, \sqrt{2}\} \subseteq \mathbb{R}$

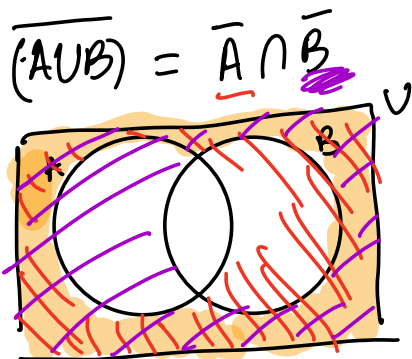
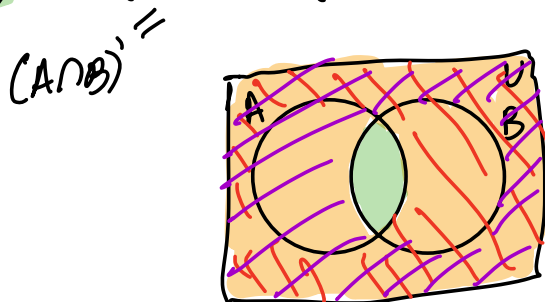
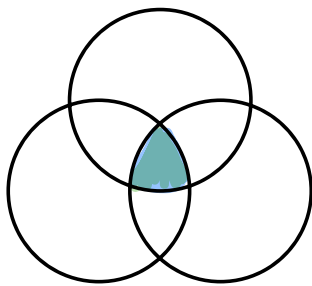
$C = \{i, -i\} \subseteq \mathbb{C}$

cartesian product: $A \times B = \{(a, b) : a \in A, b \in B\}$.

Venn Diagram:



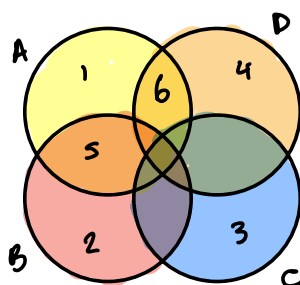
$$(A \cap B) \cap C = A \cap (B \cap C) \quad (A \cap B)^c = \overline{(A \cap B)} = \bar{A} \cup \bar{B}$$



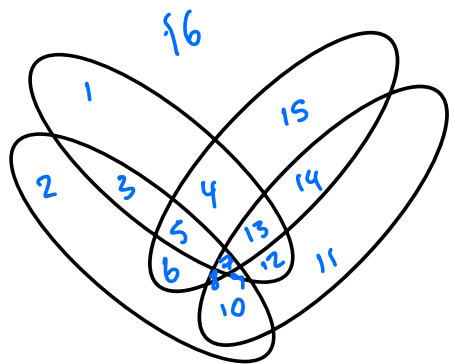
A, B, C, D :

A, B, C, D

$A \cap B, A \cap D, A \cap C$



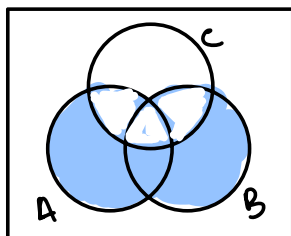
NOT A
VENN DIAGRAM



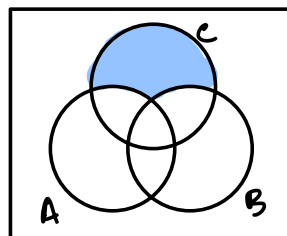
2^n regions for n sets.

Ex: Draw a Venn Diagram for A, B, C

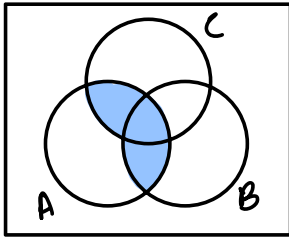
a) $(A \cup B) - C$



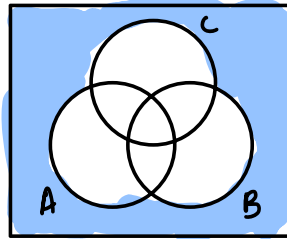
c) $\overline{(A \cup B)} \cap C$



$$b) A \cap (B \cup C)$$

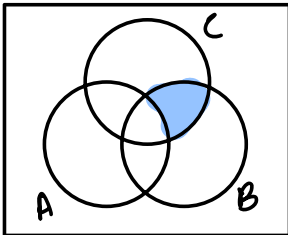


$$d) \bar{A} \cap \bar{B} \cap \bar{C} = \overline{(A \cup B \cup C)}$$



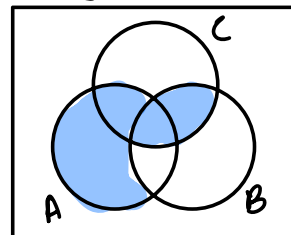
Write expressions for the shaded area using $A, B, C, \cup, \cap, \bar{}, -$.

a)



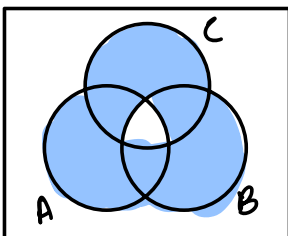
$$(B \cap C) - A$$

b)



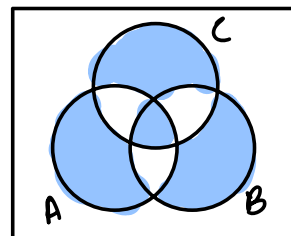
$$(A - B) \cup (B \cap C)$$

c)



$$A \cup B \cup C - A \cap B \cap C$$

d)



$$(\overline{A \cup B \cup C} - A \cap C - A \cap B - B \cap C) \cup (A \cap B \cap C)$$

3. True or False:

$$1) \overline{(A \cap B)} = \bar{A} \cup \bar{B}$$

True

$$2) \overline{A \cup B} = \bar{A} \cap \bar{B}$$

True

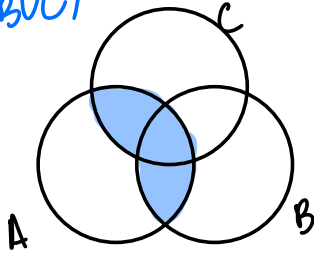
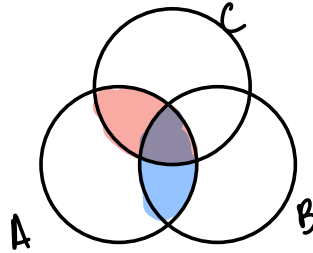
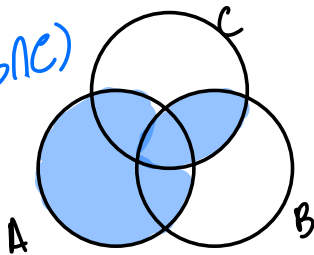
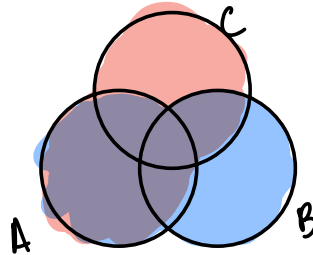
{ De Morgan's Law

$$3) A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

~~False?~~ True. Distribution.

$$4) A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

True.

$A \cap (B \cup C)$  $A \cap B \cup A \cap C$  $A \cup (B \cap C)$  $A \cup B \cap A \cup C$ 

Let A be a set.

if $|A| = n$

$$|\mathcal{P}(A)| = |\{X : X \subseteq A\}| = 2^n$$

$\mathcal{P}(A)$ is the power set of A .

$$\mathcal{P}(\{0, 1\}) = \{\{0\}, \{1\}, \{0, 1\}, \emptyset\}$$

$$\mathcal{P}(\{a, b, c\}) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$

$$\mathcal{P}(\mathbb{N}) = \left\{ \emptyset, \mathbb{N}, \{1\}, \{2\}, \{3\}, \{p : p \text{ is prime}\}, \{2n : n \in \mathbb{N}\}, \{3n : n \in \mathbb{N}\}, \dots, \{2n+1 : n \in \mathbb{N}\}, \dots \right\}$$

$$|\mathbb{N}| = \aleph_0 \quad \text{aleph}$$

Fact: $|\mathcal{P}(A)| = 2^{|A|}$ when $|A| = n$.

Ex. 1. $\mathcal{P}(\{3, 0, 1, 2\}) = \{ \emptyset, \{2\}, \{3, 0, 1\}, \{3, 0, 1, 2\} \}$

2. $\mathcal{P}(\{\mathbb{N}, \mathbb{R}\}) = \{ \emptyset, \mathbb{N}, \mathbb{R}, \{\mathbb{N}, \mathbb{R}\} \}$

~~3. $\mathcal{P}(\{X : X \subseteq \{1, 2, 3\}, 3 \notin X\})$~~

3. $\{X \in \mathcal{P}(\{1, 2, 3\}) : 3 \notin X\}$
 $= \{ \emptyset, \{1\}, \{2\}, \{1, 2\} \}$

True or False

a) $\mathcal{P}(1) = \{ \emptyset, \{1\} \}$? No, 1 is not a set.

b) $\mathcal{P}(\{3, 1, 2\}) = \{ \emptyset, \{1\}, \{2\}, \{1, 2\} \}$ No.

$|\mathcal{P}(\{3, 1, 2\})| = 2^3 = 2^{|\{3, 1, 2\}|}$

$A = \{1, 2\} \quad \mathcal{P}(A) = \{X : X \subseteq A\}$

$$\boxed{A \notin \{A\} \quad ; \quad A \in \{A\}}$$

Let A, B with $|A| = n$, $|B| = m$.

$$a) |\mathcal{P}(\mathcal{P}(\mathcal{P}(A)))| = 2^{2^n}.$$

$$b) |\mathcal{P}(A \times B)| = 2^{|A||B|} = 2^{n \cdot m}$$

$|A \times B| = |A||B|.$

$$c) |\mathcal{P}(A) \times \mathcal{P}(B)| = |\mathcal{P}(A)| |\mathcal{P}(B)| = 2^n \cdot 2^m = 2^{n+m}.$$

$$d) \left| \left\{ X : X \in \mathcal{P}(A) \text{ and } |X| \leq 1 \right\} \right| = |A| + 1.$$

||

$$= n + 1.$$

$$\left| \{ \emptyset, \{a_1\}, \dots, \{a_n\} \} \right|$$

$$e) |\mathcal{P}(A \times \emptyset)| = |\mathcal{P}(\emptyset)| = 2^{|\emptyset|} = 1.$$

$$\mathcal{P}(\emptyset) = \{ \emptyset \}.$$

Indexed sets.

$$A_1, B_1, C, \dots = A_1, A_2, A_3, \dots$$

def. let I be a set. then if $\{A_i\}_{i \in I}$ are indexed sets, then

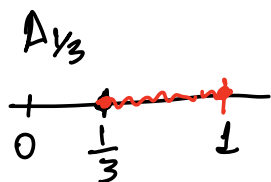
$$\bigcap_{i \in I} A_i = \{x : x \in A_i \text{ for every } i \in I\}$$

$$\bigcup_{i \in I} A_i = \{x : x \in A_i \text{ for at least one } i \in I\}$$

a) $I = \{1, 2, \dots, n\}$

$$\bigcap_{i \in I} A_i = A_1 \cap A_2 \cap \dots \cap A_n.$$

b) if $I = [0, 1]$ and $A_i = [i, 1]$



$$\bigcap_{i \in I} A_i = \bigcap_{i \in [0, 1]} [i, 1] = \{1\}$$



Note that $A_1 = [1, 1] = \{1\}$.

c) $I = \mathbb{N}$, $A_n = \{n, n+1, n+2, \dots\}$

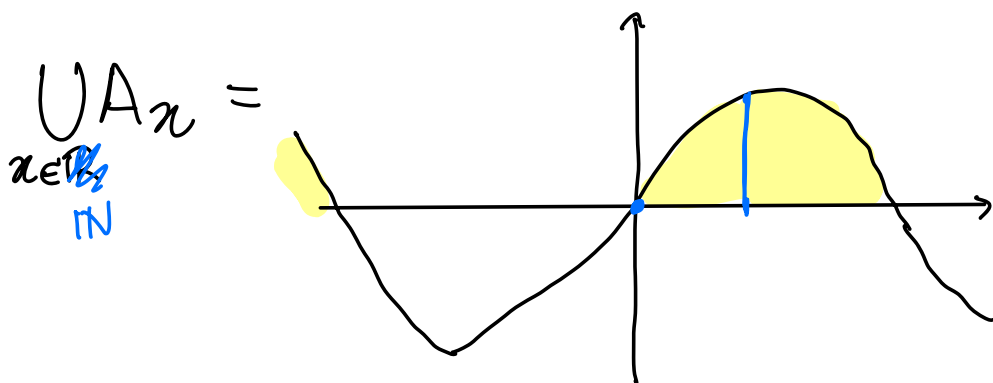
$$\bigcap_{n \in \mathbb{N}} A_n = \{1, 2, 3, \dots\} \cap \{2, 3, 4, \dots\} \cap \{3, 4, 5, \dots\} \cap \dots$$

$$= \emptyset$$

$$2023 \notin A_{2024}$$

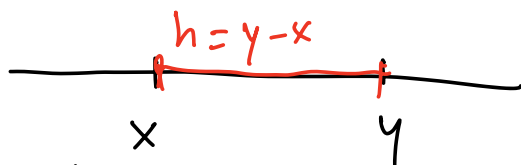
$$n \notin A_{n+1}$$

d) $I = \mathbb{R}$, $A_x = \{ (x, y) \in \mathbb{R}^2 : 0 \leq y \leq \sin(x) \}$.



Relations: $x \leq y$

$$y = x + h \quad \text{for some } h \geq 0.$$



$$A = [-2023, 1000] ; \text{ minimal element: } -2023.$$

instead $A \subseteq \mathbb{N}$. is there a min element?

$$A = \{ m, m+h_1, m+h_2, \dots \}$$

Fact: Well-ordering Principle.

Ex: $A = \left\{ \frac{1}{n} : \underline{n \in \mathbb{N}} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\}$

$$a, b \in \mathbb{Z}; \quad a = qb + r, \quad q \in \mathbb{Z}, \quad 0 \leq r < b. \\ r \in \mathbb{Z}.$$

$$A = \{ a - xb : x \in \mathbb{Z}, 0 \leq a - xb \} \subseteq \mathbb{N}$$

so by the Well-ordering Principle, there is a min
element of A .

$$\downarrow \\ a - qb = r \Rightarrow a = qb + r$$

$b=3$

$$A_0 = \{ \overset{r=0}{qb+0} : q \in \mathbb{N} \} = \{ 3, 6, 9, 12, \dots \}$$

$$A_1 = \{ \overset{r=1}{qb+1} : q \in \mathbb{N} \} = \{ 1, 4, 7, \dots \}$$

$$A_2 = \{ \overset{r=2}{qb+2} : q \in \mathbb{N} \} = \{ 2, 5, 8, \dots \}$$

$$Q: A_0 \cup A_1 \cup A_2 = \mathbb{N}.$$

Follow up. Given $b \in \mathbb{N}$, what is

$$A_0 \cup A_1 \cup \dots \cup A_{b-1} = \mathbb{N}.$$

$$\text{for } A_i = \{ qb + i : q \in \mathbb{N} \}.$$

Punch-line: Given $n \in \mathbb{N}$, you can decompose $\mathbb{N}$

by the A_r 's where r is remainder of

EUCLIDIAN

EUCLIDES.

$$k = qn + r, \quad k \in \mathbb{Z}, k \geq 0, \\ q \in \mathbb{Z}, q \geq 0.$$

euclidian division
integer division.

$$r \in \mathbb{Z}, \\ 0 \leq r < n$$

$$\text{complement: } A^c = \overline{A}$$

Axioms : ZFC.

Zermelo-Fraenkel - Choice.

8 axioms

Axiom Choice.